

NAME: _____

PERIOD: _____

DATE: _____

One Table, Two Study Designs

AP Statistics · Unit 3 · Topic 3.14 · B: 2026-12-22 · E: 2027-02-05

[STAR] TODAY'S PROBLEM

Suppose a transit authority obtains the shown counts under either design.

Design A: Take independent SRSs of 100 from each of three route populations of 4,000 and record ticket method.

Design B: Take one SRS of 300 from 15,000 system riders and record both route and ticket method.

Ari: “Route rows imply homogeneity; the counts are enough to choose.”

Bea: “Read how riders were sampled; identical counts can support different population claims.”

Ticket method

Route	Mobile	Card	Cash	Total
North	60	25	15	100
East	48	32	20	100
West	42	33	25	100
Total	150	90	60	300

FIRST TAKE — before the video (5 min)

Which claim is more defensible? Cite the sampling structure of each design.

[BOOK] LET DATA COLLECTION NAME THE TEST (keep on the page)

Use chi-square **homogeneity** to compare one categorical distribution across independently sampled populations or randomized treatments: H_0 says the distributions are the same. Use chi-square **independence** for one random sample from one population with two categorical variables recorded: H_0 says the variables are not associated. Both use $E = (\text{row total})(\text{column total})/(\text{table total})$, require every expected count > 5 , and have $df = (r - 1)(c - 1)$. The arithmetic may match even when the inferential wording does not.

Finish after the video:

DOK 1 (a) For each design, state the number of random samples and the categorical variable or variables recorded.

DOK 2 (b) Calculate the expected counts and df , then verify random sampling, 10 percent, and expected counts for both designs.

DOK 3 (c) Adjudicate the claims. For each design, name the procedure and state H_0 and H_a in context. Use the sampling structure, diagnose the rejected table-only rule, and identify the numerical pieces that remain identical.

Frame: For Design A, the procedure is _____ because _____; for Design B, it is _____ because _____.

Frame: The rejected rule would make readers assume _____; the numerical pieces that match are _____.

EXIT — turn this in

One thing the video changed about my first take: _____